ModelIt!

Student Activity Pack

How Can One Mutation Change Everything?

Modeling Gene Mutations, Protein Structure, and Phenotypic Variation

Grade 12L1 | NGSS HS-LS3-2 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is a mutation? Is it always a bad thing? Explain your thinking with any examples you know.

2 How do you think a change in DNA can affect the way your body looks or functions?

3 If every human has millions of mutations compared to each other, why aren't we all sick? What does that suggest about most mutations?

4 Have you heard of sickle cell disease or cystic fibrosis? What do you know about what causes them at the genetic level?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Mutation Type | Gene Location | Protein Structure Change | Protein Function | Phenotypic Impact

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Silent Change

Mutation Type: Point (synonymous) | Gene Location: Coding region, non-critical site | Protein: Hemoglobin

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sickle Cell Mutation

Mutation Type: Point (missense, E6V) | Gene Location: HBB gene, position 6 | Protein: Hemoglobin beta chain

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Frameshift Disaster

Mutation Type: Single base insertion | Gene Location: Early in CFTR gene coding region | Protein: CFTR chloride channel

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Epigenetic Modification — Chemical changes to DNA or histone proteins that alter gene expression without changing the DNA sequence — can amplify or suppress the phenotypic impact of a mutation by changing how much of the mutated protein is actually produced

Genetic Background — The combined effect of all other genes in an individual's genome — modifier genes can enhance or suppress the phenotypic impact of a specific mutation, explaining why the same mutation causes severe disease in one person and mild symptoms in another

Environmental Interaction — External factors like diet, temperature, UV exposure, and chemical exposure that interact with genotype to determine phenotype — some mutations are only harmful under specific environmental conditions, revealing gene-environment interactions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

A single change in one letter of your DNA — out of 3.2 billion — can cause sickle cell disease, give you red hair, protect you from HIV, or increase your cancer risk. How can one tiny mutation in a gene have such massive effects on an organism, and why do some mutations help while others harm? Use evidence from your simulation to support your answers.

1. **How does your model explain why a single amino acid change can sometimes cause fatal disease and sometimes have no effect at all?**

2. **Why is the redundancy of the genetic code (multiple codons for the same amino acid) an important buffer against mutation damage?**

3. **Your model shows that frameshift mutations are almost always devastating. Why is the reading frame so important for protein function?**

4. **If most mutations are neutral, why do people associate the word 'mutation' with disease and danger? How does your model challenge that assumption?**

5. **How does genetic variation from mutations connect to evolution by natural selection? Why are mutations essential for adaptation?**

8 STEM Challenge: Design a Genetic Screening Decision Tool

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YOUR ENGINEERING MISSION

Design a decision-support tool that helps genetic counselors explain to patients what their specific gene mutations mean — distinguishing between harmful, neutral, and potentially beneficial variants using the protein structure-function model.

THE SCENARIO

A genetics clinic receives hundreds of patient genome sequences each week, each containing thousands of mutations compared to the reference genome. Most mutations are harmless, but a few may cause disease. Genetic counselors need a tool that uses the principles from your model — mutation type, gene location, protein structure change, and functional impact — to rapidly classify mutations and explain results to patients in understandable terms.

Engineering Design Thinking

1. WHAT SPECIFIC MODEL-BASED CRITERIA DOES YOUR TOOL USE TO CLASSIFY MUTATIONS AS HARMFUL, NEUTRAL, OR POTENTIALLY BENEFICIAL?

2. HOW DOES YOUR TOOL COMMUNICATE COMPLEX GENETIC INFORMATION TO PATIENTS WITHOUT SCIENTIFIC TRAINING?

3. WHAT MODEL EVIDENCE SUPPORTS YOUR CLASSIFICATION CRITERIA, AND HOW DOES YOUR TOOL HANDLE UNCERTAIN OR AMBIGUOUS MUTATIONS?

4. HOW DOES YOUR TOOL ACCOUNT FOR THE FACT THAT PHENOTYPIC IMPACT CAN VARY BETWEEN INDIVIDUALS CARRYING THE SAME MUTATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Genetic counselors help patients understand genetic test results and make health decisions, earning $80,000-$110,000/year. Bioinformaticians who analyze genome sequences and classify mutation variants earn $75,000-$140,000/year. Structural biologists studying protein folding and mutation effects work in pharmaceutical and biotech companies, earning $80,000-$160,000/year.